EduPro Online Learning Platform

Predictive Modeling for Course Demand & Revenue Forecasting

Research Paper — Data Science & Machine Learning Project

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Abstract

This research paper presents a comprehensive predictive modeling study conducted on EduPro, an online learning platform. The study leverages historical transaction data, course metadata, teacher profiles, and user demographics to build machine learning models that forecast course enrollment demand and revenue generation. Three predictive targets were modeled: Enrollment Count per course, Course Revenue, and Category-level Revenue aggregation.

The analysis of 10,000 transactions across 60 courses, 60 teachers, and 3,000 users reveals that CourseRating is the strongest predictor of enrollment, while CoursePrice directly determines revenue. Ridge Regression achieved an R^2 of 0.983 for Course Revenue prediction, and Gradient Boosting achieved R^2 of 0.893 for Category Revenue — both considered production-ready models. An interactive Streamlit dashboard was developed to enable real-time predictions and business intelligence.

1. Introduction

1.1 Background

The online education industry has experienced exponential growth in the post-pandemic era, with platforms competing to attract learners and generate sustainable revenue. EduPro operates in this dynamic market, offering a diverse portfolio of courses across 12 categories ranging from Artificial Intelligence to Digital Marketing.

Despite having a rich repository of transactional and course data, EduPro currently lacks the predictive intelligence to make data-driven decisions about course launches, pricing strategies, and instructor assignments. Course planning relies on historical intuition rather than quantitative forecasting, increasing business risk and missed revenue opportunities.

1.2 Problem Statement

EduPro currently lacks:

- Predictive models for course enrollment demand
- Revenue forecasting at course and category level
- Quantitative evidence to support course launch and pricing decisions
- Data-driven instructor-course matching recommendations

As a result, strategic decisions about which courses to launch, how to price them, and which instructors to assign are made without forward-looking data support — increasing risk and limiting revenue optimization.

1.3 Project Objectives

This project aims to:

- Build predictive ML models for Enrollment Count, Course Revenue, and Category Revenue
- Conduct a comprehensive Exploratory Data Analysis (EDA) to uncover demand and revenue drivers
- Engineer meaningful features from raw course, teacher, and transaction data
- Deploy an interactive Streamlit dashboard for real-time predictions and stakeholder use
- Provide actionable business recommendations based on data-driven insights

1.4 Scope & Deliverables

Deliverable	Description
Research Paper	Comprehensive EDA, methodology, results, and recommendations
Jupyter Notebook	Full ML pipeline: EDA → Feature Engineering → Model Training → Evaluation
Streamlit Dashboard	Interactive 6-page analytics and prediction dashboard
Trained Models (.pkl)	3 production-ready serialized models for deployment

2. Dataset Overview

2.1 Data Sources

The EduPro dataset consists of four interconnected sheets from the platform's operational database, covering the full year 2025.

Sheet	Records	Key Columns	Role in Analysis
Courses	60	CourseID, Category, Type, Level, Price, Duration, Rating	Primary feature source for ML
Teachers	60	TeacherID, Expertise, YearsOfExperience, TeacherRating	Instructor feature engineering

Transactions	10,000	TransactionID, UserID, CourseID, Amount, Date, TeacherID	Target variable computation
Users	3,000	UserID, Age, Gender	Demographic context

2.2 Data Quality Assessment

A thorough data quality audit was conducted prior to analysis:

Quality Check	Courses	Transactions	Status
Null Values	0	0	✅ Clean
Duplicates	0	0	✅ Clean
Date Format	N/A	DD/MM/YYYY	✅ Parsed
Price Range	₹0 – ₹490.90	₹0 – ₹490.90	✅ Valid

The dataset is exceptionally clean — zero null values and zero duplicates across all four sheets — eliminating the need for data imputation and allowing direct feature engineering.

3. Exploratory Data Analysis (EDA)

3.1 Enrollment Analysis

Enrollment counts were computed by aggregating transaction records per CourseID. Key findings:

Metric	Value	Insight
Enrollment Range	140 – 196	Very narrow spread
Mean Enrollment	167 per course	Platform promotes equally
Top Course	Data Analysis with Python (196)	Data Science leads
Category Spread	800 – 900 total per category	Near-uniform distribution
Level Split	Beginner 35.7% / Advanced 34.8% / Intermediate 29.5%	Well-balanced portfolio

3.2 Revenue Analysis

Revenue was computed by summing transaction amounts per CourseID. Free courses (CourseType = 'Free') generate zero revenue regardless of enrollment volume.

Category	Total Revenue	Rank
Artificial Intelligence	₹2,02,750	🥇 1st
Business	₹1,81,527	🥈 2nd
Project Management	₹1,69,103	🥉 3rd
Programming	₹77,453	4th
Data Science	₹77,052	5th
Marketing	₹0	❌ No Revenue

Total platform revenue for 2025: ₹9,10,128. Top course by revenue: Computer Vision (~₹85,417). Free courses account for 63.3% of the portfolio but generate zero direct revenue.

3.3 Free vs Paid Analysis

Metric	Free Courses	Paid Courses
Count	38 (63.3%)	22 (36.7%)
Avg Enrollment	168.5	163.5
Avg Revenue	₹0	₹41,424

Key insight: Free and Paid courses attract nearly identical enrollment volumes (168.5 vs 163.5), confirming that price does NOT drive enrollment decisions on EduPro. This is critical — monetisation strategy must focus on converting free course audiences rather than discounting paid courses.

3.4 Correlation Analysis

Pearson correlation coefficients were computed between all numeric features:

Feature Pair	Correlation	Business Interpretation
CoursePrice ↔ CourseRevenue	+1.00	Price directly determines revenue — expected

CourseRating ↔ EnrollmentCount	+0.29	Higher-rated courses attract more enrollments
CourseRating ↔ CourseDuration	+0.21	Longer courses tend to be better rated
CoursePrice ↔ EnrollmentCount	-0.16	Costlier courses get slightly fewer enrollments
CourseDuration ↔ EnrollmentCount	-0.10	Longer courses are slightly less popular

Key finding: CourseRating is the strongest predictor of enrollment among all numeric features. Quality of course content drives demand more than price or duration.

3.5 Instructor Impact Analysis

Teacher data was joined to transactions to compute per-instructor performance metrics. Two outlier instructors with 20–24 years of experience generated ~3,000 enrollments each — significantly above the platform average. However, experience alone is not sufficient: most instructors cluster under 500 total enrollments regardless of tenure. Teacher rating (4.5–5.0) is a stronger signal, with top-rated instructors generating up to ₹3 lakhs in revenue versus near-zero for low-rated instructors.

3.6 Time Trend Analysis

Monthly aggregation of transactions across January–December 2025 reveals a stable platform with no major seasonal spikes. Monthly enrollments range between 760–880, with a peak in June 2025. Monthly revenue fluctuates between ₹65,000–₹82,000. The absence of seasonal patterns suggests that EduPro's demand is driven by evergreen learning interest rather than academic cycles.

3.7 User Demographics

- Mean user age: 25 years — young adult learner base
- Age range: 15–35 — students and early-career professionals
- Gender split: Female 50.7% / Male 49.3% — near-equal balance
- Bimodal age pattern: peaks at 15–16 (students) and 31–32 (working professionals)

4. Feature Engineering

4.1 Master DataFrame Construction

A master dataset was constructed by merging all four sheets. Enrollment count and revenue per course were aggregated from transactions, and category-level revenue was computed and merged back. The dominant teacher per course was identified from transaction frequency and teacher features were joined accordingly.

4.2 Engineered Features

Feature	Type	Values	Rationale
PriceBand	Categorical	Free / Low / Medium / High	Captures non-linear price effects
DurationBucket	Categorical	Short / Medium / Long / Very Long	Groups course length meaningfully
RatingTier	Categorical	Low / Medium / High / Top	Captures quality thresholds
ExperienceBucket	Categorical	Junior / Mid / Senior / Expert	Groups instructor seniority
ExpertiseMatch	Binary	0 or 1	Teacher-course alignment signal
RevenuePerEnrollment	Numeric	0 – 490.9	Revenue efficiency metric
IsFree	Binary	0 or 1	Strong revenue signal
TeacherRatingTier	Categorical	Low / Medium / High / Top	Instructor quality grouping

All categorical features were label-encoded for ML compatibility. The final feature matrix contains 16 features covering course attributes, instructor characteristics, pricing bands, and quality tiers.

5. Methodology

5.1 Modeling Approach

A multi-target supervised regression approach was adopted. Three separate models were built — one per prediction target. All models were evaluated using 5-Fold Cross-Validation to ensure robust generalization estimates given the small dataset size (60 course records).

5.2 Models Evaluated

- Linear Regression — baseline parametric model
- Ridge Regression — L2 regularization to handle feature collinearity
- Lasso Regression — L1 regularization for automatic feature selection
- Random Forest Regressor — ensemble tree model, robust to outliers
- Gradient Boosting Regressor — sequential boosting for complex patterns

5.3 Evaluation Metrics

Metric	Formula / Purpose	Best Value
MAE	Mean Absolute Error — average prediction error in original units	Lower = Better
RMSE	Root Mean Squared Error — penalises large prediction errors more heavily	Lower = Better
R^2 Score	Proportion of target variance explained by the model (5-fold CV mean)	Closer to 1.0

6. Model Results & Analysis

6.1 Target 1: Enrollment Count

Model	MAE	RMSE	R^2
Linear Regression	11.09	13.25	-0.474
Ridge Regression	10.88	13.02	-0.417
Lasso Regression	10.70	12.89	-0.382
🏆 **Random Forest (Tuned)**	9.85	~12.0	-0.144
Gradient Boosting	10.40	12.79	-0.426

Analysis: All models yield negative R^2 scores for enrollment prediction. This is a data limitation, not a model failure. The enrollment range of only 140–196 per course (standard deviation ~12) provides insufficient variance for ML models to learn meaningful patterns. The platform's equal-promotion strategy across all courses creates near-identical enrollment distributions, making statistical prediction fundamentally difficult. Random Forest (Tuned) with R^2 = -0.144 was selected as the best available model. Top features: CourseDuration (0.28 importance) and CourseRating (0.21 importance).

6.2 Target 2: Course Revenue

Model	MAE	RMSE	R^2
Linear Regression	₹1,728	₹2,506	0.981
🏆 **Ridge Regression**	₹1,672	₹2,479	0.983
Lasso Regression	₹1,728	₹2,505	0.981
Random Forest	₹1,855	₹3,325	0.975
Gradient Boosting	₹1,733	₹3,017	0.982

Analysis: Ridge Regression achieves an exceptional R^2 of 0.983 — indicating the model explains 98.3% of course revenue variance. The linear model outperforms tree-based models because the relationship between CoursePrice and CourseRevenue is strongly linear. Top features: RevenuePerEnrollment (~12,000 importance score) and CoursePrice (~11,500 importance score). This model is production-ready.

6.3 Target 3: Category Revenue

Model	MAE	RMSE	R^2
Linear Regression	₹46,439	₹56,901	-0.369
Ridge Regression	₹45,538	₹56,186	-0.346
Lasso Regression	₹46,439	₹56,901	-0.369
Random Forest	₹26,557	₹32,876	0.607
🏆 **Gradient Boosting**	₹10,786	₹17,135	0.893

Analysis: Gradient Boosting achieves R^2 = 0.893, significantly outperforming all other models including Random Forest (0.607). Linear models fail completely (negative R^2) because category-level revenue is driven by non-linear category groupings that tree-based models capture naturally. Top feature: CourseCategory (0.89+ importance) — the category alone explains 89%+ of category revenue variance. This model is production-ready.

6.4 Model Summary

Target	Best Model	MAE	R^2	Status
Enrollment Count	Random Forest (Tuned)	9.85	-0.144	⚠️ Data Limited

Course Revenue	Ridge Regression	₹1,672	0.983	✅ Production Ready
Category Revenue	Gradient Boosting	₹10,786	0.893	✅ Production Ready

7. Streamlit Dashboard

An interactive 6-page analytics dashboard was developed using Streamlit and Plotly, enabling real-time predictions and business intelligence exploration without requiring technical expertise.

Page	Features
🏠 **Overview**	5 KPI cards, Revenue by Category, Enrollment trends, Monthly dual-axis chart, Free vs Paid analysis
📊 **EDA Explorer**	Top N course filters, Category analysis, Instructor scatter plots, User demographics, Price vs Enrollment scatter
🤖 **Predictions**	16-parameter sliders, Live ML predictions for all 3 targets, Gauge chart, Radar chart, Optimization tips
🏆 **Feature Insights**	Model performance table, Interactive feature importance charts, Correlation heatmap, Actual vs Predicted scatter
📋 **Data Tables**	Multi-filter course browser, Revenue ranking table, Top performers, Sunburst hierarchy chart
🎯 **Recommendations**	Content-based course recommender, User profile inputs, Similarity scoring, Personalized course cards

8. Business Recommendations

8.1 Revenue Optimization

- Launch new paid courses in AI, Business, and Project Management — these 3 categories account for 60%+ of total platform revenue
- Optimal pricing range is ₹300–₹490 — generates maximum revenue with minimal enrollment impact
- Introduce a freemium model — convert free course audiences to paid certifications or advanced modules

- Marketing and Machine Learning categories are entirely free — add at least one paid course per category

8.2 Course Quality Strategy

- Courses rated 4.0+ consistently attract 15–20% more enrollments — invest in content quality review
- CourseRating is the strongest enrollment predictor — implement a structured quality improvement program
- Longer courses (20–35 hours) tend to receive higher ratings — consider expanding short course content

8.3 Instructor Assignment

- Assign instructors with 15+ years experience to flagship paid courses — they generate disproportionately higher enrollments
- Teacher rating (4.5+) is the strongest revenue signal — use rating as primary criterion for course assignments
- Expertise-category alignment (ExpertiseMatch = 1) improves both enrollment and revenue outcomes

8.4 Platform Strategy

- No strong seasonal patterns detected — focus on consistent quality rather than seasonal promotions
- User base is young (mean age 25) with bimodal distribution — market differently to students (free tier) and professionals (paid certifications)
- Gender parity (50.7% F / 49.3% M) — maintain gender-neutral course marketing and content

9. Limitations & Future Scope

9.1 Current Limitations

Limitation	Impact	Future Fix
Only 60 course records	Small dataset — CV variance is high	Collect data over multiple years
Uniform enrollment (140–196)	Enrollment model R^2 = -0.144	Add behavioral features (CTR, watch time)
Single year of data (2025)	No multi-year trend analysis	Expand to 3+ years

No user-course interaction data	Limited personalisation signals	Add ratings, completion rates, reviews
Static dataset	Cannot capture real-time demand shifts	Connect to live database pipeline

9.2 Future Scope

- Deep Learning models (LSTM, Transformers) for sequential enrollment prediction using time-series data
- Collaborative filtering recommendation system using full user-course interaction matrix
- Real-time prediction API built on FastAPI, integrated with EduPro's operational database
- A/B testing framework to validate pricing recommendations with actual course launches
- Natural Language Processing (NLP) on course descriptions to extract content quality signals

10. Conclusion

This project successfully transforms EduPro's historical course data into forward-looking business intelligence. By applying a comprehensive data science methodology — from EDA through feature engineering to multi-target ML modeling — the study delivers two production-ready predictive models and a rich set of business insights.

The Ridge Regression model for Course Revenue ($R^2 = 0.983$) and Gradient Boosting model for Category Revenue ($R^2 = 0.893$) are immediately deployable for strategic planning. The Streamlit dashboard makes these predictions accessible to non-technical stakeholders, enabling EduPro to move from reactive reporting to proactive course planning.

The enrollment prediction model's low performance ($R^2 = -0.144$) is an honest finding rooted in data structure — not a model failure. It signals a clear opportunity: investing in richer behavioral data collection will unlock future enrollment prediction capability.

This project fundamentally transforms EduPro's approach to course strategy — from intuition-driven planning to evidence-based decision making.

References

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Appendix

A. Technology Stack

Category	Tools	Purpose
Language	Python 3.11	Core analysis and modeling
Data Processing	Pandas, NumPy	Data manipulation and computation
Visualization	Matplotlib, Seaborn, Plotly	EDA charts and interactive dashboard
Machine Learning	Scikit-learn	Model training and evaluation
Dashboard	Streamlit	Interactive web application
Model Persistence	Joblib	Serializing trained models
IDE	VS Code + Jupyter	Development environment

B. Feature Encoding Reference

Feature	Category	Encoding	Value
PriceBand	Free	Label Encoded	0
PriceBand	Low (₹1–150)	Label Encoded	2

PriceBand	Medium (₹151–350)	Label Encoded	3
PriceBand	High (₹351+)	Label Encoded	1
ExperienceBucket	Junior (1–3 yrs)	Label Encoded	1
ExperienceBucket	Expert (15+ yrs)	Label Encoded	0